Fertilizer in Two Greenhouses

Unit 1 · Topic 1.13 · TODAY'S PROBLEM

Suppose a grower has 24 similar tomato plants, 12 in a sunny greenhouse and 12 in a shaded greenhouse. The response is fruit yield in kilograms per plant.

Design A: Completely randomize all 24 plants; assign 12 to fertilizer and 12 to no fertilizer, ignoring greenhouse.

Design B: Treat greenhouse as a block; within each greenhouse, randomly assign 6 plants to fertilizer and 6 to no fertilizer.

Pilot means from plants grown under the same conditions are shown.

Mean fruit yield (kg per plant)

Greenhouse	Fertilizer	None
Sunny	7.2	6.0
Shaded	4.0	2.8

FIRST TAKE (5 min, on your sheet)

Which design would you use to detect the fertilizer effect most clearly? Name one detail behind your choice.

DOK 1 **(a)** Name the blocking variable and the response variable, with units.

DOK 2 **(b)** For the sunny greenhouse, find fertilizer minus no fertilizer. For the fertilized plants, find sunny minus shaded. Compare the two differences.

DOK 3 **(c)** Which design should the grower use? Use the two differences to explain your choice and how chance imbalance could affect Design A. If Design B gives a statistically significant difference, state a causal conclusion limited to the plants and conditions studied.

Video + follow-along: `u3_lesson6-7_live.html` (scan the code)
Finish (a)–(c) after the video. Turn in your sheet.